

MEDORA 7 E, ND, USA

WMO: 727520

Lat: 46.8946N

Lon: 103.3769W

Elev: 845

StdP: 91.58

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94080

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.8	-23.1	-29.3	0.3	-25.8	-26.5	0.4	-23.2	8.1	-0.8	7.4	-2.5	2.5	N/A	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	33.4	18.3	31.3	17.8	29.4	17.6	20.8	28.1	19.8	27.3	18.8	26.5	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	14.9	24.4	17.3	13.7	23.2	16.1	12.7	21.7	64.4	27.8	60.4	27.3	56.7	26.5	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.6	5.9	DB	-29.7	36.7	2.2	2.3	-31.3	38.3	-32.6	39.7	-33.9	41.0	-35.5	42.6	(4)
(5)			WB	-30.0	23.2	2.2	2.0	-31.5	24.6	-32.8	25.8	-34.0	26.9	-35.6	28.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-7.2	-7.8	-0.3	5.5	11.5	17.3	21.6	20.3	15.1	6.8	-0.1	-7.1	(6)	
	DBStd	12.01	7.99	7.88	7.25	5.64	4.73	3.78	3.35	3.58	5.00	5.44	6.71	7.75	(7)	
	HDD10.0	2528	534	499	324	152	39	2	0	0	13	127	307	530	(8)	
	HDD18.3	4615	793	733	578	384	214	61	8	22	122	357	554	788	(9)	
	CDD10.0	1207	0	0	5	17	87	221	361	319	164	29	3	0	(10)	
	CDD18.3	251	0	0	0	0	4	29	111	82	23	1	0	0	(11)	
	CDH23.3	2968	0	0	1	11	67	323	1232	974	337	23	0	0	(12)	
	CDH26.7	1148	0	0	0	2	11	89	522	400	118	6	0	0	(13)	
(14) Wind	WSAvg	3.1	3.3	3.0	3.3	3.4	3.2	3.0	2.7	2.7	2.8	3.1	3.0	3.1	(14)	
(15) Precipitation	PrecAvg	389	12	11	19	35	59	82	56	31	32	30	14	10	(15)	
	PrecMax	507	34	40	52	92	114	141	150	76	75	107	74	29	(16)	
	PrecMin	256	2	1	4	11	18	28	16	3	4	5	0	1	(17)	
	PrecStd	60	7	8	12	20	32	30	32	16	21	29	16	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	12.1	21.3	24.8	28.2	32.4	36.7	35.7	33.0	26.0	19.2	10.7	(19)	
		MCWB	4.4	5.6	9.3	12.0	14.8	18.8	18.7	17.4	18.0	13.6	9.5	4.7	(20)	
	2%	DB	7.1	7.9	16.9	20.5	25.2	29.1	33.8	33.0	30.0	21.7	14.8	7.2	(21)	
		MCWB	2.8	3.6	7.6	10.0	14.8	18.0	18.7	17.9	16.4	12.3	7.5	2.7	(22)	
	5%	DB	4.9	5.3	13.6	17.7	22.9	26.8	31.6	31.0	27.6	18.3	11.7	5.0	(23)	
		MCWB	1.5	1.8	6.2	8.6	13.5	17.1	18.5	17.3	15.1	10.2	5.5	1.3	(24)	
	10%	DB	3.0	2.8	10.1	15.1	20.5	24.9	29.7	28.8	24.5	15.5	9.0	3.0	(25)	
		MCWB	0.3	0.3	4.7	7.3	12.4	16.1	18.7	17.1	14.6	8.9	4.1	0.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	6.6	10.5	12.8	18.1	21.9	23.7	21.2	19.6	15.0	9.7	5.4	(27)	
		MCDB	9.5	11.6	20.0	24.1	23.5	28.7	28.9	28.8	27.0	23.6	18.6	10.5	(28)	
	2%	WB	2.8	4.1	8.4	10.4	16.3	19.9	21.5	20.0	18.0	13.5	7.5	3.1	(29)	
		MCDB	6.7	8.4	16.6	18.4	22.4	26.5	27.5	27.5	25.9	20.7	14.1	6.9	(30)	
	5%	WB	1.5	1.9	7.0	8.9	15.0	18.2	20.3	18.9	16.5	11.1	5.6	1.6	(31)	
		MCDB	4.4	5.1	14.2	16.1	20.8	25.2	27.9	27.0	24.9	17.4	10.6	5.0	(32)	
	10%	WB	0.4	0.6	5.3	7.5	13.4	17.0	19.2	18.0	15.2	9.4	3.9	0.2	(33)	
		MCDB	2.8	3.1	10.7	14.1	19.0	23.6	27.0	25.9	22.4	15.0	8.0	3.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.5	10.5	11.4	11.1	11.4	12.9	13.3	12.8	10.6	10.1	8.7	(35)	
		MCDBR	10.3	12.2	15.8	16.2	15.1	14.9	16.2	16.9	17.4	15.9	14.3	11.1	(36)	
	5% WB	MCWBR	6.5	7.8	8.7	8.7	7.0	7.2	5.7	6.0	7.1	8.0	8.3	7.1	(37)	
		MCWBR	9.8	11.1	15.6	15.8	12.3	13.3	13.6	14.2	15.2	14.7	13.6	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.249	0.256	0.273	0.310	0.330	0.343	0.353	0.360	0.319	0.290	0.260	0.245	(40)	
		taud	2.517	2.538	2.550	2.447	2.454	2.448	2.451	2.400	2.495	2.560	2.535	2.523	(41)	
	Ebn at Noon		876	941	967	949	935	920	906	884	898	882	852	840	(42)	
		Edh at Noon	61	74	85	104	108	109	108	108	90	72	59	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.64	2.62	3.77	4.90	5.80	6.55	6.92	5.70	4.31	2.74	1.73	1.24	(44)	
	RadStd		0.13	0.27	0.29	0.29	0.52	0.49	0.28	0.38	0.32	0.30	0.21	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page